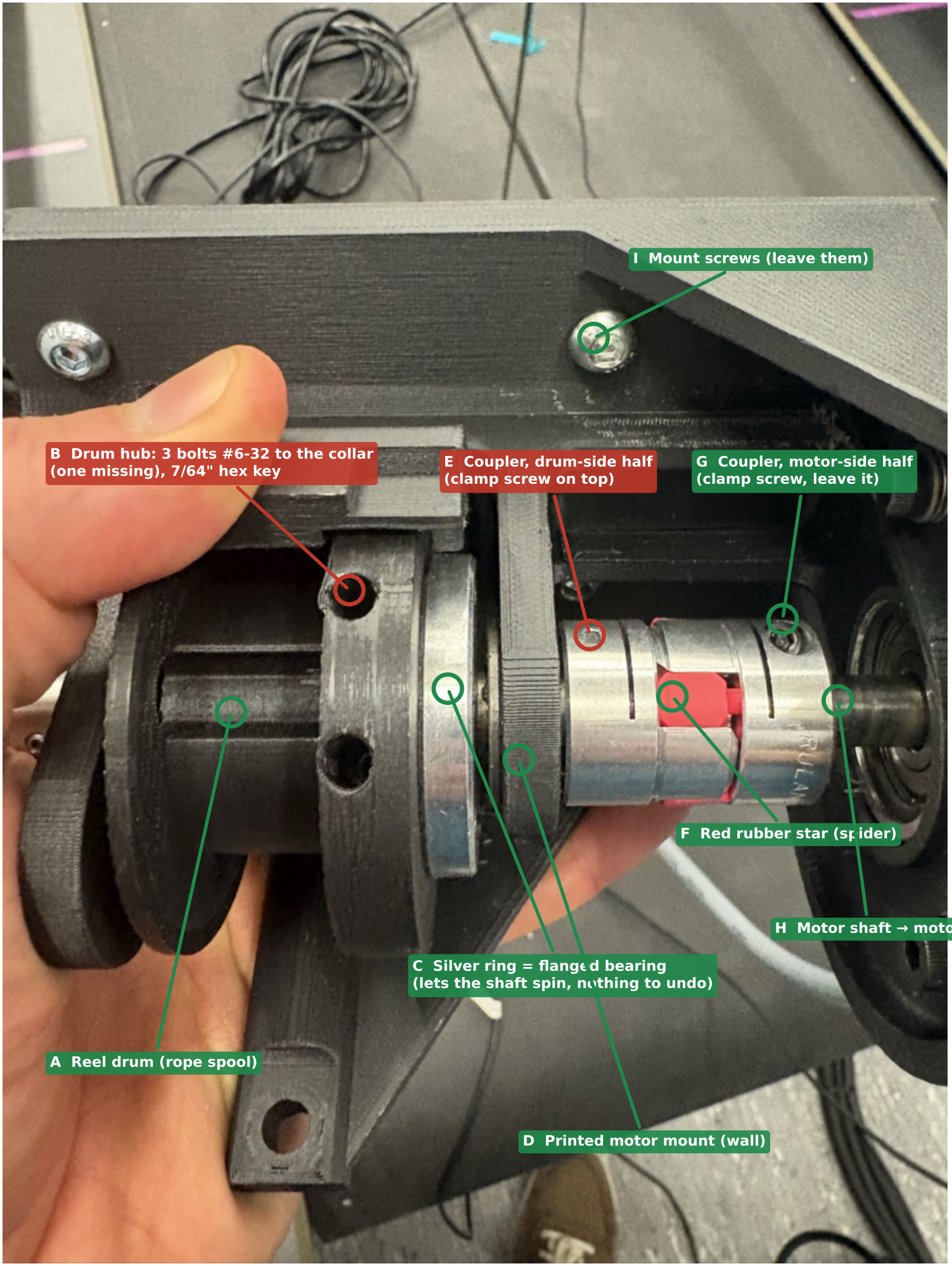
Your motor unit (tower B), part by part

Red = you work on it. Green = leave it alone.



Take the drum off

How the drum is held (Stanford build guide p. 11)

drum --3 bolts #6-32 x 1/2 in + #6 washers--> flanged shaft collar --its clamp--> 3/8 in shaft

One of the 3 drum bolts is missing. Try first: a new bolt, no taking apart (last page).

0 Prepare

- UI: Release (treadmill stopped). Stop the server. This tower's 48 V and 24 V OFF.
- Take 4 photos (front, back, both sides). Put a cup or a magnetic tray next to you.
- Marker: one line across drum hub (B), shaft and coupler (E), so it goes back the same way.

1 Free the rope

- Unhook the rope from the load cell / dummy. Unwind it from the drum.
- Keep it on the drum if you want: you re-tie it anyway (rope page).

2 Loosen the 3 drum bolts (B): 7/64 in hex key

- Turn the drum until a hole faces you; key straight down the hole; counter-clockwise.

3 Loosen the coupler's drum-side clamp screw (E)

- Small screw on top of the silver coupler half next to the black mount wall.

  Stanford's tool list: 2.5 mm hex (try 7/64 in too). Two turns are enough.
- Do NOT touch the motor-side half (G): it stays on the motor shaft.

4 Push the shaft out (3/8 in x 3 in aluminium rod, in 2 bearings)

- The drum sits between the 2 bearings: the shaft has to slide out lengthwise.
- Push one end with your thumb or a wooden/plastic rod, light taps, while holding the drum.

  If it does not go one way, try the other way.
- **It does not move with light taps? STOP and send photos of both shaft ends.**
- **Never hammer on the shaft or the printed parts.**

5 Lay the parts out in order

- Drum, collar, washers or spacers (note where they were), shaft. The red star (F) may fall

  out of the coupler: put it back into the motor-side half (G).

Fix and put it back

6 While it is apart

- Shake the drum: the missing bolt may be inside. Get it out.
- Buy 3 new #6-32 x 1/2 in socket head cap screws + 3 #6 washers (inch screws; or bring one old bolt to the store as a sample).
- Collar set screw (comes with the collar, McMaster 9793T12): file a FLAT on the aluminium shaft where it presses (shaft in a vise, ~10 mm long, ~0.5 mm deep; easy, the shaft is soft). Clean shaft, collar and screw with alcohol; new set screw if its tip is worn.
- Re-tie the rope now, while the drum is off (next page).

7 Put it back (reverse order)

- Drum + collar: 3 bolts with washers, a drop of blue threadlocker, tighten firmly (7/64 in).
- Shaft through bearing, drum/collar and the other bearing, into the coupler half (E), as deep as before (marker line). Keep ~1 mm gap at the red star: do not squeeze the claws.
- Collar set screw: threadlocker, onto the FLAT, tighten hard. Coupler clamp screw (E): firmly.
- Turn the drum by hand: it must not rub the mount or the cover.

8 Test (after 10 min for the threadlocker)

- New marker line across drum, collar, shaft and coupler.
- Rope on: at least 3 turns stay on the drum when tight.
- 24 V + 48 V on, testing page: 10 N (does it wind IN?), then 30, 50, 100 N, 1000 ms each.
- Line still straight and the pulls get stronger = fixed. Then check towers A, C, D too.

Without taking it apart (try this first)

- Put a new #6-32 x 1/2 in bolt + washer into the empty hole (on the end of the 7/64 in key), threadlocker, tighten all 3 bolts.

Labels on page 1 are read from one photo of tower B.
If your tower looks different, send a photo and the guide is updated.

Re-tie the rope on the drum

From Stanford's build guide (docs/reference/stanford-bumpem-build-guide.pdf p. 12, video "Motor Unit | Reel Drum Rope Assembly" on biomechatronics.stanford.edu/bump-em). Easiest with the drum OFF the shaft.

1 Length

- Rope length = W + 71 cm (28 in). W = how far the rope has to travel (e.g. the treadmill width).
- At least 3 turns must stay on the drum when the rope is fully out.

2 Through the tie hole

- Push the rope end from the OUTSIDE of the drum core into the small tie hole, so the end comes out inside the core (picture below). Pull ~15 cm through.

3 Stopper knot (figure 8) on that end (pictures below)

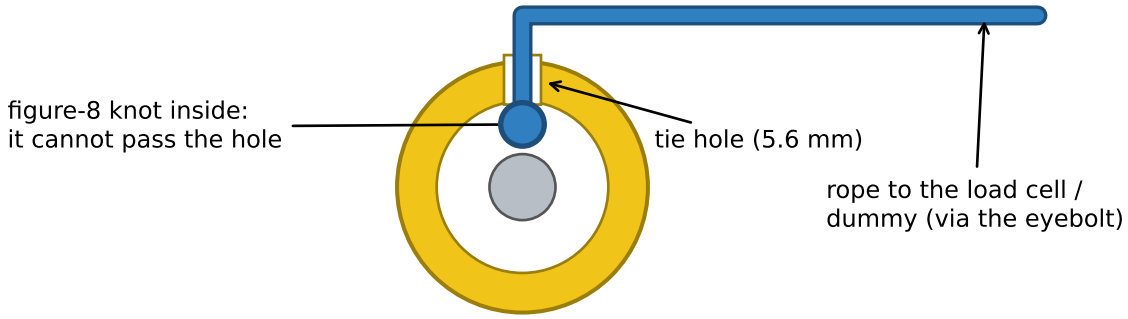
- Tie it close to the end, pull it tight, cut the tail to ~1 cm.
- Pull the rope back: the knot jams inside and cannot come through the hole.

4 Wind it on

- Drum back on the shaft (steps 7 on the previous page), then wind the rope on neatly side by side, and out through the eyebolt in the cover.
- Direction: do one 10 N test pull with the rope slack. The motor must WIND IT IN. If it unwinds instead, take the turns off and wind them the other way round.

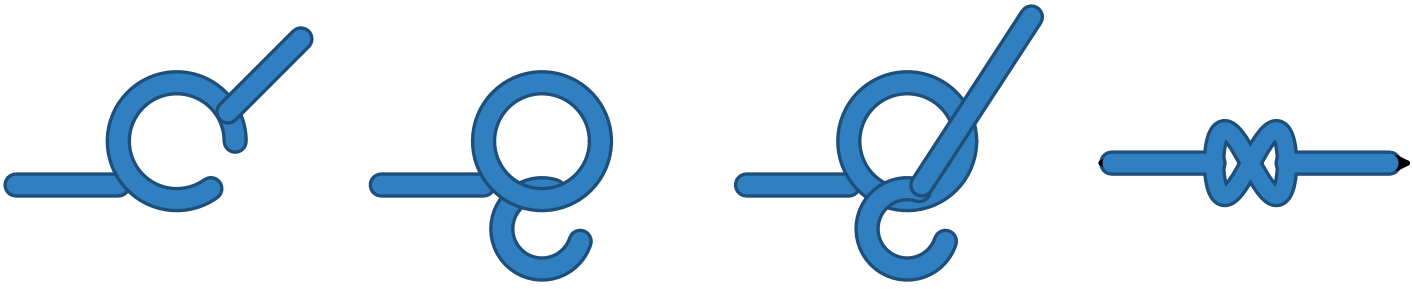
5 Other end

- Stanford ties an overhand noose (a simple knot around the rope itself) to connect to the breakaway cable / harness. Use the same connection as the other towers.



drum, cut through the tie hole

Figure-8 stopper knot (good animations: search "figure eight stopper knot")



1 loop: lay the end over the rope

2 end around the rope, behind it

3 end down through the first loop

4 pull both sides: an "8" = done